

Yihan (Chrystal) Xie

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EDUCATION

Western University

09/2024 - 08/2025 (Expected)

Master of Computer Engineering in Software Engineering

London, ON

Courses: Cloud Computing, Database Management, Machine Learning, Data Analytics, Introduction to Hacking

University of Toronto

09/2019 - 06/2024

Honors Bachelor of Science in Computer Science

Toronto, ON

Courses: Web Programming, Introduction to Software Engineering, Introduction to Database, Data Structures and Analysis, Computer Network

SKILLS

Languages: Java, Python, JavaScript, Kotlin, Golang, SQL, C

Databases & Tools: Git & Github, Docker, GCP, AWS, Firebase, MongoDB, PostgreSQL, MySQL

Technologies: React, React Native, Node.js, Express.js, Django, jQuery, RESTful, Spring Boot, Bootstrap, HTML&CSS

PROFESSIONAL EXPERIENCE

TD Bank

Toronto, ON

Business Insights & Analytics Co-op/Intern

01/2024 - 05/2024

- Provided analytics and CAATTS (Computer-Assisted Audit Tools and Techniques) support to the Global Audit Division.
- Developed Python scripts using Pandas and Numpy to replace legacy systems like ACL (Audit Command Language), ensuring timely deliverables for SQL Server and Oracle database access audits with a tight deadline on deliverables being decommissioned.
- Automated extractive text summarization on Large PDF files using Python library using Summa and Sumy.
- Build a data pipeline using Python to process and analyze large HR datasets extracted from Alteryx, improving data accuracy and enabling auditors to derive actionable insights through data cleaning, validation, segmentation, and combination.

PROJECTS

MiniSpotify: A Full-stack Android music app

01/2025 - 05/2025

- Built a Spotify-inspired music app in Kotlin by fully leveraging Android JetPack Library and Hilt Dependency Injection.
- Implemented the BottomBar and in-App Navigation using JetPack Navigation component.
- Created a mock RESTFUL API json-server and handled network requests via Retrofit.
- Built feed, album, and favourite feature by using Room Database for local persistence.
- Integrated Google Exoplayer for seamless global music playback control.

FoodEase: A Spring Boot Based online food ordering System

09/2024 - 12/2024

- Developed CRUD REST APIs using Spring Boot Controllers, encompassing functionalities such as registration, menu searching, ordering, and checkout.
- Leverage Spring Data JDBC and repositories to interface with a PostgreSQL database hosted on AWS RDS, handling data related to menus, restaurants, and uses.
- Implemented authentication via Spring Security, employing session-based authentication mechanisms.
- Employed a structured architecture (controller, service, repository) with dependency injection.
- Built a responsive frontend using ReactJS and Ant Design for cart management and orders.
- Containerized and deployed on AWS App Runner for scalability.

GenPost: An AI-Powered Social Networking Web App

09/2024 - 12/2024

- Designed and developed a fullstack social networking platform using React.js for the frontend and Go for the backend services.
- Integrated OpenAI's DALL-E 3 API to enable users to generate and customize AI-assisted post content.
- Improved the authentication using token based registration, login, and logout flow with React Router v4 and server-side user authentication with JWT.
- Deployed scalable Go-based web services on Google App Engine to handle post creation and retrieval.
- Leveraged Elasticsearch (deployed to GCE) to provide fast and flexible search features that allow users to search recent posts and list personal posts.